



Lab 08: Inter-VLAN Routing

Router1 (2911) · Switch0 (2960) · PC0 · PC1 · PC2 · PC3

Lab #08

Topic: Inter-VLAN Routing

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Extend Lab 07 by adding Router1 (2911) to enable inter-VLAN routing.
- Configure trunk link from Switch0 Gig0/1 to Router1 Gig0/0.
- Set up 802.1Q sub-interfaces on Router1 — one gateway per VLAN.
- Assign PC0–PC3 to four separate VLANs on Switch0.
- Verify full communication between all four VLANs using ping.

2. Network Topology

Router1 (2911) is on the left, connected via Gig0/0 to Switch0 (2960-24TT) on Gig0/1 — this is the trunk link. PC0 on Fa0/1 (VLAN 10), PC1 on Fa0/2 (VLAN 20), PC2 on Fa0/3 (VLAN 30), PC3 on Fa0/4 (VLAN 40). Router1 sub-interfaces act as the gateway for each VLAN.

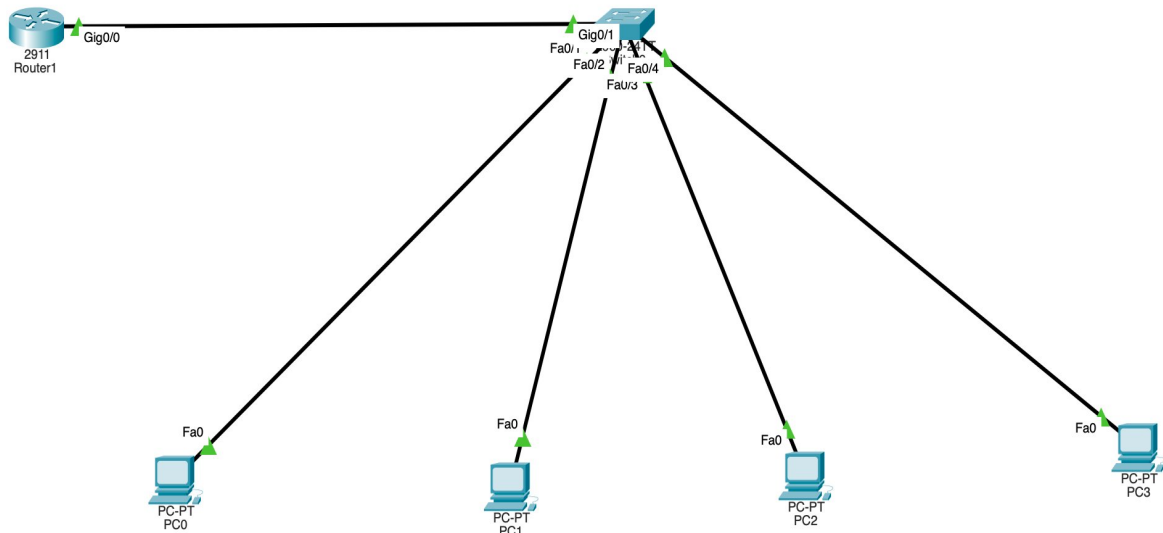


Figure 1 – Router1 (2911) trunk to Switch0; PC0–PC3 on VLANs 10/20/30/40

3. IP Addressing Table

Device	Interface/Port	VLAN	IP Address	Subnet Mask	Gateway
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Router1	Gig0/0.10	10	192.168.10.1	255.255.255.0	—
Router1	Gig0/0.20	20	192.168.20.1	255.255.255.0	—
Router1	Gig0/0.30	30	192.168.30.1	255.255.255.0	—
Router1	Gig0/0.40	40	192.168.40.1	255.255.255.0	—
PC0	Fa0/1 (Switch)	10	192.168.10.10	255.255.255.0	192.168.10.1
PC1	Fa0/2 (Switch)	20	192.168.20.10	255.255.255.0	192.168.20.1
PC2	Fa0/3 (Switch)	30	192.168.30.10	255.255.255.0	192.168.30.1
PC3	Fa0/4 (Switch)	40	192.168.40.10	255.255.255.0	192.168.40.1

4. Step-by-Step Configuration

4.1 Switch0 – VLANs, Access Ports & Trunk

```
Switch(config)#vlan 10
Switch(config-vlan)#name VLAN10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name VLAN30
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name VLAN40
Switch(config-vlan)#exit

Switch(config)#interface Fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface Fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface Fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface Fa0/4
Switch(config-if)#switchport mode access
```



```
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit

Switch(config)#interface GigabitEthernet0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch#write memory
```

4.2 Router1 – Sub-interfaces

```
Router1(config)#interface GigabitEthernet0/0
Router1(config-if)#no shutdown
Router1(config-if)#exit

Router1(config)#interface GigabitEthernet0/0.10
Router1(config-subif)#encapsulation dot1Q 10
Router1(config-subif)#ip address 192.168.10.1 255.255.255.0
Router1(config-subif)#exit

Router1(config)#interface GigabitEthernet0/0.20
Router1(config-subif)#encapsulation dot1Q 20
Router1(config-subif)#ip address 192.168.20.1 255.255.255.0
Router1(config-subif)#exit

Router1(config)#interface GigabitEthernet0/0.30
Router1(config-subif)#encapsulation dot1Q 30
Router1(config-subif)#ip address 192.168.30.1 255.255.255.0
Router1(config-subif)#exit

Router1(config)#interface GigabitEthernet0/0.40
Router1(config-subif)#encapsulation dot1Q 40
Router1(config-subif)#ip address 192.168.40.1 255.255.255.0
Router1(config-subif)#exit
Router1#write memory
```

5. Verification

```
PC0>ping 192.168.20.10    ! VLAN10 to VLAN20 – expect Reply
PC0>ping 192.168.30.10    ! VLAN10 to VLAN30 – expect Reply
PC0>ping 192.168.40.10    ! VLAN10 to VLAN40 – expect Reply

Router1#show ip route      ! 4 connected subnets via sub-interfaces
Switch#show interfaces trunk ! Gig0/1 must show as trunk, all VLANs allowed
```

6. Key Takeaways

- Router-on-a-Stick uses ONE physical trunk link to route between all VLANs — cost-effective.



- The parent interface (Gig0/0) must be 'no shutdown' — all sub-interfaces depend on it.
- encapsulation dot1Q tags outgoing frames with the correct VLAN ID for the switch to process.
- Four VLANs = four sub-interfaces = four separate /24 subnets all routed by Router1.
- A Layer 3 switch performs inter-VLAN routing in hardware — faster and more scalable.